

Improving Energy Efficiency through Green Cloud Computing in IoT Networks

Aishwarya Shekhar

SCSE, Galgotias University Greater Noida, India
aishwaryashekhar26@gmail.com

Abdul Aleem

SCSE, Galgotias University, Greater Noida, India
abdul.aleem@galgotiasuniversity.edu.in

Abstract— Improving the energy efficiency by using green cloud computing cover through an IoT network which involves optimizing the resource utilization and implementing all the smart technologies. By integrating the sensors and the actuators, the IoT network can dynamically adjust the power consumption based on demand, which leads to a more sustainable and efficient type of cloud infrastructure. Additionally, using the machine learning algorithms can forecast workload patterns, allowing for energy management. These new research contribute to a greener cloud environment, reducing both the operational costs and environmental impact. The integration of Renewable Energy in Cloud Computing explore ways to integrate renewable energy sources into cloud data centers, ensuring a more sustainable, recyclable and eco-friendly cloud infrastructure. Dynamic Resource Allocation in Cloud Computing with IoT investigate how IoT devices can assist in real-time resource allocation, optimizing energy usage and enhancing overall efficiency in cloud environments. Carbon Footprint Reduction through Cloud-Based IoT Solutions discusses strategies to reduce the footprint of carbon coming from IoT devices and cloud services, promoting environmentally conscious practices in the digital realm. The present study aims to investigate the potential synergies between cloud computing and smart grid technologies in order to establish a more integrated and energy-efficient environment. By processing the data closer to the source and limiting the need for lengthy data transfers, edge computing can help reduce the energy usage of cloud services. This is explored in Edge Computing for Green Clouds. Blockchain for Sustainable Cloud Computing examines how block chain technology might improve cloud services' accountability and transparency while encouraging environmentally friendly behavior. Low-Power IoT Devices for Cloud Connectivity explores the creation and application of low-power IoT devices that promote a greener IoT ecosystem by enabling cloud connectivity while consuming the least amount of energy possible.

Keywords—Green Cloud Computing, Green Servers, IoT Networks, Green IoT

I. INTRODUCTION

The environmental effect of data centers is becoming more challenging concern as demand for cloud computing services rises. The Green cloud computing has come out as a major field for research and development, with the goal of lowering energy and carbon footprints. The amalgamation of Internet of Things (IoT) networks into cloud infrastructure is one area that shows promise for advancement. By working together, we hope to improve monitoring capabilities, maximize resource

usage, and create a more sustainable computing environment. Because cloud computing is dynamic, resource allocation must be done effectively. Real-time server load and energy usage observation is made possible by the amalgamation of IoT sensors into data centers. By permitting dynamic resource allocation algorithms, this data assures optimal utilization of computing resources. Subsequently, servers use less energy overall as they adapt in real-time to shifting workloads. Smart cooling systems come up with an additional level of energy efficiency by using the IoT-connected sensors. By collecting information on temperature and humidity inside data centers, these sensors permit precise control over cooling systems. With the help of this knowledge, cooling systems may adjust to the distinctive requirements of the hardware and saves energy for the betterment of ecosystem. This intelligent cooling process reduces environmental impact while improving overall energy efficiency to consume less power. Resource necessities are predicted in large part by predictive analytics using the machine learning algorithms, using data provided by the Internet of Things sensors and actuators. These algorithms predicts peak consumption times by examining the past usage trends. This forecast prevents oversupply during low-usage periods by authorizing proactive actions to effectively control changing demand, like as resource scaling and load balancing [1].

Energy efficient data centers, servers and data storage are given first priority by the help of algorithms that consider energy consumption of these IoT devices. These algorithms bear in mind the energy efficiency of various parts inside the data centers by utilizing IoT data. Workloads are being routed to the most energy-efficient resources, reducing the overall energy usage without relinquishing the performance and dependability that cloud computing services are known for. Amalgamating renewable or recyclable energy is essential for a green cloud's infrastructure. Energy generation from the renewable sources like solar, water or wind power must be tracked by IoT devices from time to time. A sustainable or eco-friendly cloud ecosystem is facilitated by the data center operations that are in line with the accessibility of renewable energy, which also decreases the bad impact of computing on the environment. These days edge computing used by IoT devices brings processing closer to the data source. This method reduces the need for extensive data transmit to centralized cloud servers. By utilizing energy-efficient processors and exploring hardware design innovations, data centers can further improve their overall energy efficiency [2].

Efficient communication protocols play a very important role in the green Internet of Things. This includes the development of lightweight communication protocols and efficient data collection techniques. In addition, the green Internet of Things includes the responsible management of data generated by IoT devices. This comprise applying edge computing solutions to process the data closer to the source, minimizing the need for extensive data transmission and centralized processing. By minimizing unnecessary data transmission, the green Internet of Things promotes energy efficiency and reduces environmental impact. Security and privacy considerations are important green IoT practices [3].

II. BACKGROUND

A. Green Cloud Computing

Green cloud computing is a justifiable approach to cloud technology that gives attention on minimizing the environmental effect of processing, storing, utilizing and managing data in the cloud servers. This eco-friendly paradigm focuses on energy saving, resource efficiency and the deployment and integration of renewable energy sources in cloud infrastructures. One of the main parts of green cloud technology is the optimization of data centers and servers, which are the backbone of all the cloud services like IAAS, PAAS, and SAAS. Data servers always consume significant amounts of power to cool and run the servers from time to time. Green cloud computing projects include implementing the energy-efficient hardware, advanced servers cooling systems, and intelligent resource allocation algorithms of machine learning to improve overall efficiency and reduce the energy consumption on day to day basis. These techniques minimize the carbon footprint and carbon emission and promote a greener eco-friendly cloud ecosystem. Another important part of green cloud is the use of renewable or recyclable energy sources to power data centers. These days cloud service providers are increasingly adopting solar, wind, water and other sustainable energy solutions to meet their energy needs. The transition to reduce the energy consumption not only aligns with global sustainability goals, but also ensures cloud services running on environmentally friendly devices. In addition to this, virtualization plays a key role in green cloud services. By virtualizing the servers and pooling the resources, cloud service providers can optimize hardware uses, which will reduce the uses physical servers and thus energy consumption will be done. This method not only increases the overall efficiency of cloud operations, but it can also promotes a more sustainable, more eco-friendly and cost-effective cloud infrastructure. The green cloud computing presents a forward-looking approach to all the types of cloud technology that prioritizes environmental sustainability and reduces carbon emission [4]. Strategies and technologies used in green cloud computing are the energy-efficient data centers which uses advanced cooling systems as well as optimized server hardware and renewable energy sources to reduce energy and power consumption.

The process of virtualization Combines several virtual servers into a single physical server so as to improve the

utilization of resources and to reduce the number of physical servers required on day to day basis. The dynamic resource allocation increase or decrease the cloud resources on demand to minimize energy waste during periods of low usage. Green Data Center Locations choose data centers with access to renewable energy sources and cooler climates to reduce cooling costs. Green cloud computing is good for the environment, but it can also lead to cost savings for businesses by reducing energy consumption and improving resource utilization [5].

B. Green IoT Networks

Green IoT Networks deals with Internet of Things (IoT) infrastructure that focuses on environmental sustainability and energy efficiency. Green IoT networks are an emerging concept in the Internet of Things (IoT) field, which addresses environmental issues linked with traditional IoT infrastructures. The main focus of these networks are on reducing energy consumption that means less use of power, promoting sustainability and eco-friendly hardware and minimizing the overall carbon footprint which are developed by the carbon emission coming from the old devices which use more power. The main strategies in developing green IoT networks are to reduce the energy use of IoT devices. This optimization includes the research and development of energy-efficient hardware and the implementation of energy-saving algorithms using machine learning technique. We can extend the battery life of devices and reducing the energy consumption during the use. Green IoT networks helps in maintaining the creation of a more sustainable and eco-friendly friendly IoT ecosystem. The implementation of renewable or recyclable energy sources in IoT infrastructure is the main part of green IoT networks. Amalgamating solar, wind, water or other renewable energy solutions can decrease the dependence on traditional energy sources and minimize the environmental impact of IoT implementation. In addition to recyclable hardware and energy aspects, the design and implementation of communication protocols play a pivotal role in achieving green IoT goals. Adding subsequently the energy-efficient reusable hardware, renewable energy sources and communication protocols, these networks find out the way to a more sustainable and eco-friendly friendly future in the rapidly evolving Internet of Things era. The green IoT network is very much essential to increase the energy efficiency which is being required in today's scenario to reduce the carbon emission and carbon footprints which is disturbing the ecological balance and causing green house effect [6].

C. Green Internet of Things

Green IoT the short form of Green Internet of Things refers to the applications of the IoT (Internet of Things) principles and technologies to create environmentally friendly or eco-friendly solutions. This simply means that using the IoT devices and systems to monitor, manage and to reduce the environmental impact of different processes and activities. The aim of green IoT is to make industry more energy efficient, reducing the wastage and mitigating the environmental impact of technology which cause green house effect[7].

Green IoT devices can be used to optimize energy consumption in transportation, buildings and industrial processes. As we know that, smart thermostats can adjust cooling and heating systems based on usage and weather conditions to save power consumption. Green IoT sensors can be used to gather information about the air quality, water quality and other environmental factors of the ecosystem. This information which we get through the IoT devices can be used for early detection of pollution or more economical management of resources for improving the energy efficiency. Green IoT can also be applied to waste management, enabling smarter recycling and waste collection. For example, smart bins can signal when they need to be emptied, which reduces the fuel consumption of garbage trucks. IoT devices can be used in smart agriculture to optimize the crop management, conserve water, forecasting weather and to reduce the use of pesticides and fertilizers, used for promoting sustainable agricultural practices. IoT-based intelligent transportation systems can reduce the traffic congestion, and improve overall mobility through solutions such as traffic management, EV charging infrastructure and autonomous vehicles used in day to day life.

The goal is to reduce resource consumption and minimize environmental impacts. Green IoT can promote more sustainable and environmentally friendly practices in various sectors such as agriculture, transportation, industry and smart cities. This often involves the use of sensors, data analytics and automation to optimize resource use and promote green decision-making [8].

D. Improving Energy Efficiency through Green Cloud

Improving the efficiency via green cloud computing refers to the practice of implementing, designing and using the recyclable green cloud computing resources and services in an environmentally sustainable and energy-efficient recyclable manner. The goal is to reduce the environmental impact of data centers and cloud infrastructure, which are known to consume significant amounts of energy and produce carbon emissions [9]. The green cloud will use recyclable hardware or reusable device which will surely helps in increasing the energy efficiency in today's scenario

Some strategies and technologies used in green cloud computing are:

- Energy-efficient data centers: Using advanced cooling systems, optimized server hardware, and renewable energy sources to reduce energy consumption.
- Dynamic resource allocation: Scaling cloud resources up or down based on demand to minimize energy waste during periods of low usage.
- Green or recyclable data center locations: Choosing data center locations with access to renewable energy sources and cooler climates to reduce cooling costs.
- Energy-efficient hardware: Using energy-efficient processors, storage devices, and networking equipment in data centers.
- Cloud resource management: Implementing policies and tools to efficiently allocate and de-allocate resources, preventing idle or underutilized servers.

Green cloud computing not only benefits the environment but can also lead to cost savings for businesses through reduced energy consumption and improved resource utilization.

E. Improving the Energy Efficiency through Green IoT

Low-power components, energy-efficient microcontroller's chips, and recyclable hardware play a very important role in enhancing the battery life of all the IoT devices. Researches like energy harvesting, where devices capture and utilize less amount of energy, further contribute to sustainable IoT operations. Similar to the green cloud, integrating renewable energy sources into IoT deployments improves energy efficiency. Solar-powered sensors and all the solar devices reduce the reliance on traditional power sources and offer a sustainable solution for the environment, particularly in remote or off-grid locations. Improving energy efficiency in green cloud computing and IoT requires a multifaceted approach. From optimizing data center operations and virtualization in cloud computing to designing energy-efficient devices and implementing sustainable energy sources in IoT, these strategies collectively contribute to a more environmentally responsible and energy-efficient technological landscape. As technology continues to advance, the pursuit of greener computing and IoT solutions remains crucial for a sustainable future [10].

F. Digitalization for adopting Green Computing

So far as we know that digitalization is elucidated as the operation of converting the paper based details into the digital format so that it can be used online via the computer and internet. In the past decade we have seen massive digitalization. All our information's, which were previously stored in files and notebooks, are now available in digital format where we can instantly access it from our devices. People can use their phones, tablets and laptops to do things like grocery shopping and banking. The total time spent on these electronic devices has risen dramatically in recent years, as all the work and social lives are moving online. Customers who previously used to interact with companies physically or through calls now use digital devices and social media for the communication. In the era of digitalization's organizations are rapidly embracing these technologies to automate their processes and provide a better user experience to all their customers. For example, all the banks and retailers, have websites and mobile apps that allow customers to bank and shop without leaving their homes. Digitalization is more than just converting paper documents to digital formats. Our household items are now becoming digital as they are linked to the internet. We can now control these devices remotely with the help of internet [11]. There are connected devices all around us like the phones, vehicles, and household items. These connected devices are often referred to as IoT devices - the Internet of Things. With the establishment of 5G networks,

our devices are more connected than ever before and working also in a very fast and efficient manner. 5G networks are defined as the fifth generation of cellular networks. It is Up to 100 times faster than 4G as well as 3G. 5G network is creating never-before-seen opportunities for people and businesses.

III. WORK TO BE DONE ON INTERNET OF THINGS

There is a wide range of work to be done on IoT (Internet of Things) to further its development and maximize its potential. Here are some key areas where work is ongoing or needed:

- **Security:** IoT devices are often vulnerable to cyber attacks. Developing robust security measures and standards for IoT is crucial to protect data and privacy.
- **Interoperability:** IoT devices from different manufacturers often don't work seamlessly together. Efforts are underway to establish common communication protocols and standards to enhance interoperability.
- **Scalability:** IoT networks must be able to handle the increasing number of connected devices. Scaling IoT infrastructure, both in terms of hardware and software, is essential.
- **Low-Power Devices:** Creating more efficient and longer-lasting power sources for IoT devices, especially those in remote or inaccessible locations, is a priority.
- **Data Management:** IoT generates vast amounts of data. Developing efficient data storage, processing, and analytics solutions is essential to extract meaningful insights [12].
- **Edge Computing:** Processing data closer to the source (at the edge) rather than sending it all to centralized servers can reduce latency and improve IoT system efficiency.
- **Privacy:** Addressing concerns around the privacy of individuals in IoT ecosystems is crucial. Implementing robust privacy policies and practices is essential.
- **User-Friendly Interfaces:** Creating intuitive user interfaces and experiences for controlling and interacting with IoT devices is important for widespread adoption.
- **AI Integration:** Combining IoT with artificial intelligence can enhance its capabilities, but this integration needs further exploration and development.
- **Standardization:** Developing and adhering to industry standards is vital for the growth and maturity of IoT technologies.
- **Healthcare:** In the healthcare sector, there's a need for improved IoT-based remote patient monitoring and healthcare management solutions.

- **Agriculture:** Enhancing IoT applications in agriculture for precision farming, crop monitoring, and sustainable practices.
- **Smart Cities:** Expanding IoT applications for more efficient urban planning, traffic management, and resource utilization in smart cities.

These are just a few areas where work is needed in the IoT field. IoT is a rapidly evolving technology, and ongoing research, innovation, and collaboration are essential to unlock its full potential and address its challenges effectively [13].

IV. FUTURE OF GREEN IOT AND GREEN CLOUD COMPUTING

Implementing Green IoT will change our lives into a greener, healthier and Eco-friendly environment in the future which is socially and environmentally justifiable, smarter as well as safer for the ecosystem. Currently, the most thrilling areas focus on greening things such as green communication and networking, green IoT installations, green design, green services, and applications energy-saving techniques and green localisation. Energy issues such as developing low-power chipsets and energy harvesting are the main components in the development of the green IoT. Transmission of data from the sensors to the mobile cloud computing would be more useful. The WSN is amalgamated with mobile cloud via the sensor-cloud architecture [14]. This green IoT technology will help the cities to become more environmentally sustainable and eco-friendly through IoT. Moreover, a green social network as a service short form as (SNaaS) could increase the system's energy effectiveness, sensor networks, service provisioning and cloud-based WSN management. M2M technology is pivotal in saving energy and harmful emissions. For self-operating systems to work, smart machines must become smarter using machine learning. In case of traffic, machine automation delay must be reduced while taking immediate as well as appropriate action. Green design of the IoT systems that sum up to their potential as "greener" technologies is authoritative for helping organisations to meet their sustainability goals. This can be attained by giving efforts in meticulous designing and planning of IoT products and IoT projects and solutions, by using the union of remote connectivity, low-cost devices and service provisioning. Now a day's all the Research areas include development of novel as well as more efficient, compact energy storage like batteries, fuel cells and printed/polymer batteries which are eco-friendly. Also the evolution of new energy generation devices which couple energy transmission methods and energy conversion is a topic of concern these days. In order to green the IoT, it is very necessary to use less energy using devices, looking for the new resources which are reusable or recyclable, reducing the negative impacts of IoT devices on human health, and causing less environmental degradation. Then only the green IoT can greatly contribute to a sustainable, recyclable, intelligent, and green environment. On the other hand the green cloud computing is now becoming a business standard these days as it simplifies the user's accessibility also. It provides a virtual storage or saving space to the user which could be used without caring about the details of the entire mechanism of

cloud ecosystem. Here we are discussing some other reasons why every enterprise must need Green Cloud computing for their business. The Green Cloud computing eradicates the requirement of any organization to invest in storage hardware, machines and servers by their own and it also comes under cost savings schemes. Since all the services of green cloud computing will execute only over the internet, an organization does not have to worry about technical issues and other problems linked with physical storage and backup [15].

V. CONCLUSION

Leveraging green cloud computing in IoT networks presents a promising avenue for enhancing energy efficiency. By optimizing resource utilization, implementing energy-aware algorithms, and promoting sustainable practices, this approach contributes to environmental conservation and cost-effectiveness in the rapidly expanding IoT landscape. However, ongoing research and industry collaboration are vital to address challenges and refine strategies for a more sustainable and energy-efficient future [16]. Green IoT networks will surely help in reducing the carbon emission and carbon footprints which are increasing these days.

VI. FUTURE SCOPE

5G and Cloud Synergy investigate the potential of 5G networks in conjunction with green cloud computing, enabling faster and more efficient communication between IoT devices and cloud servers and ultimately improving overall system performance. Quantum Computing for Energy Optimization explore the application of quantum computing algorithms to solve complex optimization problems, providing a quantum leap in energy-efficient resource allocation and workload management within green cloud environments. Blockchain-Based Energy Trading implement blockchain technology for transparent and decentralized energy trading among various cloud data centres fostering a more sustainable and cooperative ecosystem for energy-efficient computing. Autonomous Energy-Aware Cloud Systems develop self-learning systems that leverage artificial intelligence to autonomously adapt and optimize energy consumption based on changing workloads, user patterns, and environmental factors in green cloud computing. Renewable Energy Forecasting implement advanced predictive analytics and machine learning models to forecast renewable energy availability, allowing cloud data centres to dynamically adjust their operations and maximize the utilization of green energy sources. Innovations in Cooling Technologies research and implement novel cooling technologies, such as liquid immersion cooling or advanced air-cooling systems, to further reduce the energy required for maintaining optimal temperatures in cloud data centres. Energy-Efficient Hardware Design focus on the development of energy-efficient hardware components and architectures specifically designed for green cloud computing, reducing power consumption without compromising computational performance. Regulatory Compliance and Standards investigate the establishment of international standards and regulations promoting energy efficiency in cloud computing,

encouraging adherence to sustainable practices and ensuring a global commitment to green initiatives. As the field of green cloud computing continues to evolve, these future avenues present exciting opportunities for innovation, sustainability, and enhanced energy efficiency. Continuous research and development in these areas will contribute to a more eco-friendly and resilient cloud computing landscape [17]. Removing the de-duplicated data from the cloud servers will also help in reducing the power consumption of the devices.

REFERENCES

- [1] Wenliang Mao, Zhiwei Zhao, Zheng Chang, Geyong Min, and Weifeng Gao, Energy-Efficient Industrial Internet of Things: Overview and Open Issues, 2021, IEEE.
- [2] Manoj Muniswamaiah, Tilak Agerwala and Charles C. Tappert, Green computing for Internet of Things, 2020, IEEE
- [3] Parul Goyal, Ashok Kumar Sahoo, Tarun Kumar Sharma, Pramod K. Singh, Internet of Things: Applications, security and privacy: A survey, 2020, Elsevier.
- [4] MAHMOUD A. ALBREEM, ABDUL MANAN SHEIKH, MOHAMMED H. ALSHARIF, MUZAMMIL JUSOH, AND MOHD NAJIB MOHD YASIN, Green Internet of Things (GIoT): Applications, Practices, Awareness, and Challenges, 2021, IEEE.
- [5] Abdulaziz Alarifi, Kalka Dubey, Mohammed Amoon, Torki Altameem, Fathi E. Abd El-Samie, Ayman Altameem, S. C. Sharma, Aida A. Nasr Energy-Efficient Hybrid Framework for Green Cloud Computing, 2020, IEEE.
- [6] IKRAM UD DIN, MOHSEN GUIZANI, BYUNG-SEO KIM, SUHAIDI HASSAN, AND MUHAMMAD KHURRAM KHAN, Trust management Techniques for Internet of Things, 2019, IEEE
- [7] Arshad, Rushan, et al. "Green IoT: An investigation on energy saving practices for 2020 and beyond." IEEE Access 5 (2017): 15667-15681
- [8] Chiang, Mung, and Tao Zhang. "Fog and IoT: An overview of research opportunities." IEEE Internet of Things Journal 3.6 (2016): 854-86.
- [9] Keshav Sood, Shui Yu, Dinh Duc Nha Nguyen, Yong Xiang, Bohao Feng, and Xiaoning Zhang, A Tutorial on Next Generation Heterogeneous IoT Networks and Node Authentication: IEEE Internet of Things Magazine • December 2021.
- [10] Qing Yang, Hao Wang, Xiaoxiao Wu, Taotao Wang, Shengli Zhang, and Naijin Liu, Secure Blockchain Platform for Industrial IoT with Trusted Computing Hardware: IEEE Internet of Things Magazine • December 2021.
- [11] Aishwarya Shekhar, A Very Robust Dedicated and Verified Technique by Applying the Hybridity of Cloud for Deduplication of Data: IJTRE, 2016.
- [12] Bedi, Guneet, et al. "Review of Internet of Things (IoT) in electric power and energy systems." IEEE Internet of Things Journal 5.2 (2018): 847-870.
- [13] Albreem, Mahmoud AM, et al. "Green internet of things (IoT): An overview." 2017 IEEE 4th International Conference on Smart Instrumentation, Measurement and Application (ICSIMA). IEEE, 2017.
- [14] Tahiliani, Vinita, and Mavuri Dizalwar. "Green iot systems: An energy efficient perspective." 2018 Eleventh International Conference on Contemporary Computing (IC3). IEEE, 2018.
- [15] Ning, Zhaolong, et al. "Green and sustainable cloud of things: Enabling collaborative edge computing." IEEE Communications Magazine 57.1 (2018): 72-78.
- [16] Guo, Mian, Lei Li, and Quansheng Guan. "Energy-Efficient and Delay-Guaranteed Workload Allocation in IoT-Edge-Cloud Computing Systems." IEEE Access 7 (2019): 78685-78697.
- [17] Tao, Fei, et al. "CCIoT-CMfg: cloud computing and internet of things-based cloud manufacturing service system." IEEE Transactions on Industrial Informatics 10.2 (2014): 1435-1442.